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Model Development Phase Template

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Team ID	SWTXT_20251747173488
Project Title	Productivity Prediction Web App
Maximum Marks	5 Marks

Feature Selection Report: Productivity Prediction Web App

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Introduction

The Productivity Prediction Web App uses machine learning to forecast garment worker productivity, leveraging the "Productivity Prediction of Garment Employees" dataset. This Feature Selection Report details the selection process for the dataset's features, ensuring optimal input for Linear Regression ($R^2 = 0.197$), Random Forest ($R^2 = 0.547$), and XGBoost ($R^2 = 0.482$) models. Each feature is described, with its selection status and reasoning, to enhance transparency and streamline model development.

Dataset Overview

- Dataset: "Productivity Prediction of Garment Employees"
- Source: Kaggle (https://www.kaggle.com/datasets/utkarshs/productivity-prediction-of-garment-employees)
- Size: 1197 records, 15 features (14 inputs + 1 target post-preprocessing)
- Target: `actual_productivity` (decimal, 0.2337–0.7548)
- Preprocessing Context: Missing `wip` values imputed, `department` spellings standardized, `month` derived from `date`, categorical features encoded.

Feature Selection

The following table lists each feature, its description, selection status, and reasoning for inclusion or exclusion in the ML models.

Feature ID	Feature Name	Description	Selected	Reasoning
F-1	quarter	Quarter of the year (categorical, e.g., Quarter1–Quarter5)	Yes	Captures seasonal production trends, potentially influencing productivity. Relevant for time-based analysis.
F-2	department	Department type (categorical, sewing, finishing)	Yes	Different departments have distinct productivity patterns (e.g., sewing involves more WIP). Critical for model accuracy.
F-3	day	Day of the week (categorical, e.g., Monday–Sunday)	Yes	Productivity may vary by weekday due to workforce dynamics. Provides contextual insight.
F-4	team	Team number (numerical, 1–12)	Yes	Team composition affects productivity; unique team dynamics are model-relevant.
F-5	targeted_productivity	Target productivity set by management (decimal, 0.35–0.8)	Yes	Directly correlates with actual productivity; a key driver for prediction.
F-6	smv	Standard Minute Value, time allocated for a task (decimal, 2.9–50.48)	Yes	Reflects task complexity, strongly influencing productivity. High predictive value.
F-7	wip	Work in Progress, units pending completion (integer, 0–1591)	Yes	Indicates production backlog, critical for sewing department. Imputed missing values ensure usability.
F-8	over_time	Overtime in minutes (integer, 0–25920)	Yes	Overtime impacts worker fatigue and output, a significant productivity factor.
F-9	incentive	Incentive amount in currency (integer, 0–98)	Yes	Financial incentives may boost productivity; captures motivational effects.
F-10	idle_time	Idle time in minutes (decimal, 0–6.5)	Yes	Idle periods reduce efficiency; useful for identifying productivity bottlenecks.
F-11	idle_men	Number of idle workers (integer, 0–40)	Yes	Complements `idle_time`, quantifying workforce downtime's impact.
F-12	no_of_style_change	Number of style changes in production (integer, 0–2)	Yes	Style changes disrupt workflow, affecting productivity. Low variance but relevant.
F-13	no_of_workers	Number of workers assigned (decimal, 2–59)	Yes	Worker count directly influences output capacity; a core feature.
F-14	date	Observation date (string, e.g., 1/1/2015)	No	Redundant after deriving `month`; string format is model-incompatible.
F-15	month	Month derived from `date` (numerical, 1–12)	Yes	Captures monthly trends in productivity, more concise than `date`.

Impact on Machine Learning Models

- Selecting 14 relevant features optimized model performance, with Random Forest achieving the highest R^2 score (0.547), followed by XGBoost (0.482) and Linear Regression (0.197).

- Features like `smv`, `no_of_workers`, and `targeted_productivity` were critical, showing high importance in tree-based models (Random Forest, XGBoost).

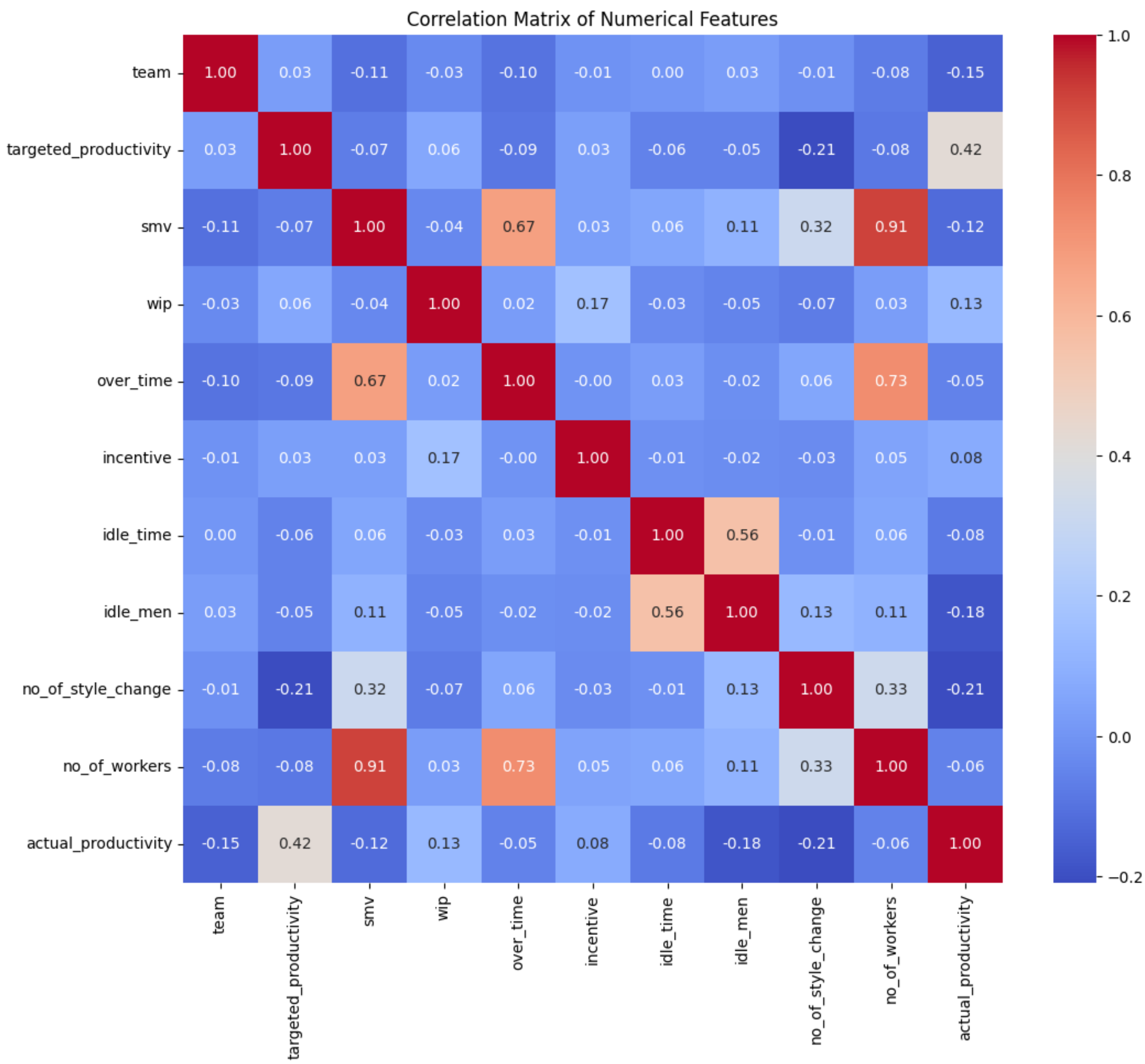
- Excluding `date` reduced redundancy, streamlining the feature set without loss of temporal information (via `month`).

- Including `wip` (post-imputation) and `department` (post-standardization) ensured data completeness, enhancing model reliability.

- The feature selection process, implemented with Pandas and Scikit-learn, directly contributed to the app's predictive accuracy.

Images

Figure : Correlation Matrix Heatmap



- Description: A heatmap visualizing correlations between numerical features (e.g., `smv`, `no_of_workers`), aiding feature selection decisions.

- Source: Generate using Seaborn (script provided below) or use a stock heatmap from Unsplash (search 'correlation heatmap').

- Placement: Insert after 'Feature Selection' section.